

Beyond property: Industrial estates and post-suburban transformation in Jakarta Metropolitan Region

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ABSTRACT

Past studies on urbanization in East Asia in general and in Indonesia in particular have taken the edge of large cities as the extension of the metropolitan core where the zones of urban–rural transition take shape. The current article argues that the emergence of global cities and decentralization trends in the last decades have challenged this traditional conception. Jakarta Metropolitan Region (JMR) is beginning to join the global trend of post-suburbia, which refers to the decentralization of urban life to the outskirts of a metropolis. Several post-suburban elements in JMR are identified in this article, including planned deconcentration of hi-tech industries and multinational companies, followed by the growth of projects beyond land and property development, including suburban cultural centers with a Western flavor. Despite these commonalities, this article argues that post-suburbia in JMR has its own uniqueness due to the reduced carrying capacity of the metropolitan core, the persistent yet relatively passive influence of the state, symbiotic relations between the market and the public sector, and the privatization of planning rules.

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Introduction

The latest debate on global urbanization increasingly pays a considerable amount of attention to post-suburbia, referring to the decentralization of urban life to the outskirts of a metropolis. Many believe that the frontier of American urban centers has moved to the edge of the metropolis (Garreau, 1991; Lang, Blakely, & Gough, 2005), forming decentralized spatial structures on the regional scale (Gordon & Richardson, 1996; Kling, Olin, & Poster, 1991a; Phelps, 2010). On a modest scale, this American trend is also emerging in Western Europe (Bontje, 2004; Borsdorf, 2004; Phelps, Parsons, & Ballas, 2006; Phelps, Wood, & Valler, 2010) and in Australia (Freestone & Murphy, 1998), where there is a more feasible role of the public sector and planning systems. Apart from these industrialized countries, several elements of post-suburbia can be found in Chinese global cities, where the local government acts as an important entrepreneur in this transformation (Wu & Phelps, 2008).

The introduction of Soeharto's market-led economy has laid the foundation for spatial transformation around Indonesia's large cities. Previous studies have shown massive rural land conversion, the creation of gated communities, and new town development at the outskirts of Jakarta, which resulted from the deregulation

and debureaucratization policies of the 1980s (Firman, 2000, 2004; Hudalah, Winarso, & Woltjer, 2007; Leaf, 1996; Leisch, 2002; Winarso & Firman, 2002). In fact, many new town developers envisioned building *kota mandiri* (autonomous towns), complete with major urban facilities and employment centers.

Although the impact of residential development on the edge of Indonesia's metropolises has been widely acknowledged, more studies need to be conducted to identify whether industrial development has played a role in creating new urban centers on the edge that are as attractive as those found in the old city centers. Following the fall of Soeharto's authoritarian and centralized regime and the commencement of the democratic and decentralized political system of the Reform Era (*Era Reformasi*) in the late 1990s, there may now be a greater possibility to creatively further transform the edge of Indonesian metropolises. More diverse actors, including the local government and the private sector, are now given more opportunities to participate in industrial development.

This article aims to explore whether urban transformation on the outskirts of Jakarta Metropolitan Region (JMR), the largest metropolitan region in Indonesia, has joined the globalizing trend of post-suburbia. It also describes the extent to which industrial development, as a leading economic sector in the region, has contributed to this transformation. In doing so, the article addresses several questions in particular. Can post-suburbia be identified in JMR? Has industrial development played a role in this transformation? What are the meanings or the unique features of post-suburbia in the context of industrializing JMR?

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Following this introduction, the article first describes post-suburban elements drawn from previous studies in the United States, Europe, and East Asia. With these conceptual elements on hand, the article outlines trends of urban deconcentration in JMR in the last decades, with a particular emphasis on the principal role of and socio-economic and political explanation for industrial investment growth in the suburbs. We then take a closer look at Kabupaten Bekasi, the industrial capital of the region. Kabupaten Bekasi was selected due to the fact that, regardless of its progressive economic restructuring in the context of an increasingly urbanizing metropolitan region, it is still administered by a rural government, potentially indicating governance complexity in a post-suburban world (Phelps et al., 2010). A more detailed analysis is conducted to point out the latest development in Jababeka, the biggest and most vibrant industrial town in Kabupaten Bekasi. The case of Jababeka indicates that industrial developers in JMR may go beyond the primary activities of selling plots and buildings to create jobs and leisure and cultural centers in the suburb. As another step of analysis, the article identifies the impacts of suburban development and provides accounts of regional governance adaptation to face this challenge. The last section discusses whether the distinctiveness of Asian extended urbanization is being eroded by this globally emerging phenomenon of post-suburbia and whether post-suburbia carries specific meanings in the context of JMR.

Post-suburbia: A global perspective

The drivers of post-suburbia

The shift from suburbia to post-suburbia that began in the developed countries was, among other reasons, triggered by the technological revolution that started at the beginning of the 20th century. First, the development of automobile technology allowed the people to travel from one edge of metropolis to another on a daily basis. Furthermore, progress in telecommunication and computer technology can potentially replace the necessity of face-to-face contact and massive physical movement, making employment decentralization more possible. As a result, since the second half of the 20th century, the American suburbs in particular have loosened their ties with their established cities. They have transformed into “outer cities” consisting of not just residential but also attractive functions such as shopping and high-tech industrial centers, which previously were an exclusive attribute of the metropolitan cores (Muller, 1982). To illustrate the spatial implication of this technological revolution, Fishman (1989) hypothesized the emergence of “technoburb,” referring to the network of dispersed urban functions influenced by the use of advanced communications and transportation technologies.

Post-suburbanization does not necessarily represent a new spatial phenomenon. As experienced by major North American cities in the first half of the 20th century, post-suburbanization can be seen as the reinvention of spatial implications of urbanization due to the production of new parameters of location decision (Lewis, 2001). In such cases, post-suburbanization did not always follow the path of new residential development but built the foundation for this residential suburbanization. Post-suburbanization was triggered by industrial development that carried new modes of production, such as the rise of industrial complexes (Hise, 2001). The creation of suburban industrial complexes was largely influenced by internal forces, including the changing behavior of and political economic alliances between developers and local leaders (Walker & Lewis, 2001).

Over the last four decades, the transformation of suburbia has been reinforced by external forces in relation to globalization of

neoliberal ideology. Suburban development is increasingly shaped by international actions and the global market. First, interregional and international competition demanded that the locational preferences of economic activities have to be more flexible. The effects of neoliberal globalization are also reflected in the people's lifestyles, such as increased individualism, symbolism, and consumerism (Kling, Olin, & Poster, 1991b). Some have argued that suburbs may adapt easier to these new global trends because they are not constrained by the past and are thus forced to innovate. Therefore, in the United States context, the new suburb is often called the “exopolis”, representing the increasing role of the exogenous forces in reshaping the city-space (Soja, 2000).

Elements of post-suburbia

Given the above context, several elements of post-suburbia, now also emerging in Europe and East Asia, can be identified. First, post-suburbia entails a shifting paradigm in the urban–rural conception. The speed of information, travel connections and globalization has made the notion of space fuzzy. As such, the suburb can no longer be seen as a transitional or privileged zone between the old city and the open countryside. Between the two contrasting realities, there can now be a new spatial form that is just as lively, attractive, and dynamic as the city. In post-suburbia, the old city and the country have dissolved into an urbanized landscape consisting of “the in between cities” (Young & Keil, 2010), or *zwischenstadt* in German (Sieverts, 2003). This “middle landscape” (Kling et al., 1991b) has major urban characteristics but is not as multifunctional as the old city center. It is no longer a cohesive spatial form where all urban activities are concentrated around a certain locational point. Instead, it consists of low-density urban spatial forms, which are structurally fragmented, economically specialized, and socially segregated (Borsdorf, 2004).

The blurring of the urban–rural conceptual divide affects the changing relations between a city and its countryside. Using the example of Orange County, California, Kling et al. (1991b) describe the emergence of the post-suburban region, referring to a decentralized spatial arrangement in which different urban activities are conducted in different places and are increasingly connected by private automobile transportation. The movement of people does not follow the concentric or radial pattern that can be found in the traditional monocentric metropolitan region. Instead, the resulting spatial patterns tend to be polycentric in nature; the suburbs gain more independency, and the centrality of the established urban core is weakened (see for example Modarres, in press).

In addition to these distinctive spatial features, post-suburbia also mirrors the shift in social relations from a communal to a libertarian orientation, favoring maximum individual freedom. In the United States, for example, the post-suburban residents tend to be more conservative and liberal on individual and policy choices such as abortion and raising taxes, respectively (Kling et al., 1991b). Moreover, Kling et al. (1991b, p. xvi) emphasized that “post-suburbanites create their lifestyle in a place where central institutions, such as the government, or the state, are less visible and functionally significant than they are in traditional urban cities”. Garreau (1991, p. xxii) termed the place where these people live as an “edge city”, partly because “the rules that govern its creation involve a search for edge – for advantage”. The devotion of edge city residents to individualism and freedom could be exemplified by the enshrined use of automobiles and luxurious private space.

The loosening role of government in the formation of the edge city may exaggerate the United States influence on post-suburbia. Western European examples show that a significant role still remains for the state in edge urban development through, for instance, city-region planning strategies and public–private

partnership schemes (Phelps et al., 2006). The resulting spatial form does not necessarily replace the traditional city center but rather complements the existing structure. Therefore instead of edge city, the European type of suburban center may be better termed “city-edge” (Bontje & Burdack, 2005). Similar results can be found in China. In fact, the Chinese central and local governments could play a more proactive role by building an ambitious planning vision and acting as key entrepreneurs and developers on the edge (Wu & Phelps, 2008).

Urban and industrial development in the Jakarta Metropolitan Region

The Jakarta Metropolitan Region (JMR) covers a total area of 5897 km². It consists of Daerah Khusus Ibukota (DKI) Jakarta or Jakarta Capital Special Province surrounded by peripheral areas known as *Bodetabek*, the acronym for Bogor–Depok–Tangerang–Bekasi (Fig. 1). Bodetabek comprises three autonomous *kabupatens* (districts or rural governments): Kabupaten Bogor, Kabupaten Tangerang, and Kabupaten Bekasi. In addition, it now also includes

five *kotas* (municipalities or urban government): Kota Bogor, Kota Depok, Kota Tangerang, Kota Tangerang Selatan, and Kota Bekasi.

In the last five decades, the population distribution of JMR has shifted considerably. This shift can be seen by comparing the results of 10-year national censuses conducted by the Central Bureau of Statistics (BPS or *Biro Pusat Statistik*, 2010) from 1961 to 2000. The shift began with the rise of metropolitan core, which occurred in the 1960s and 1970s. During this period, urbanization in JMR tended to be concentrated in the city core. More than half of the region's population increasingly resided in the main city of Jakarta. This concentration trend started to decline in the 1980s. The demographic domination of the city core was gradually overtaken by the districts and municipalities (Bodetabek). In fact, by the 1990s, the suburban population had surpassed that of the city. Although the population of the city core still grew, its rate substantially decreased. As a result, in 2000, it was predicted that the region's population had reached 22.5 million, 12.8 million of whom were sparsely distributed in Bodetabek (BPS DKI Jakarta, 2010).

The population deconcentration phenomenon might imply the transition of JMR toward a suburbanized region. Suburbanization in JMR can be reflected by the expansion of built-up areas on the

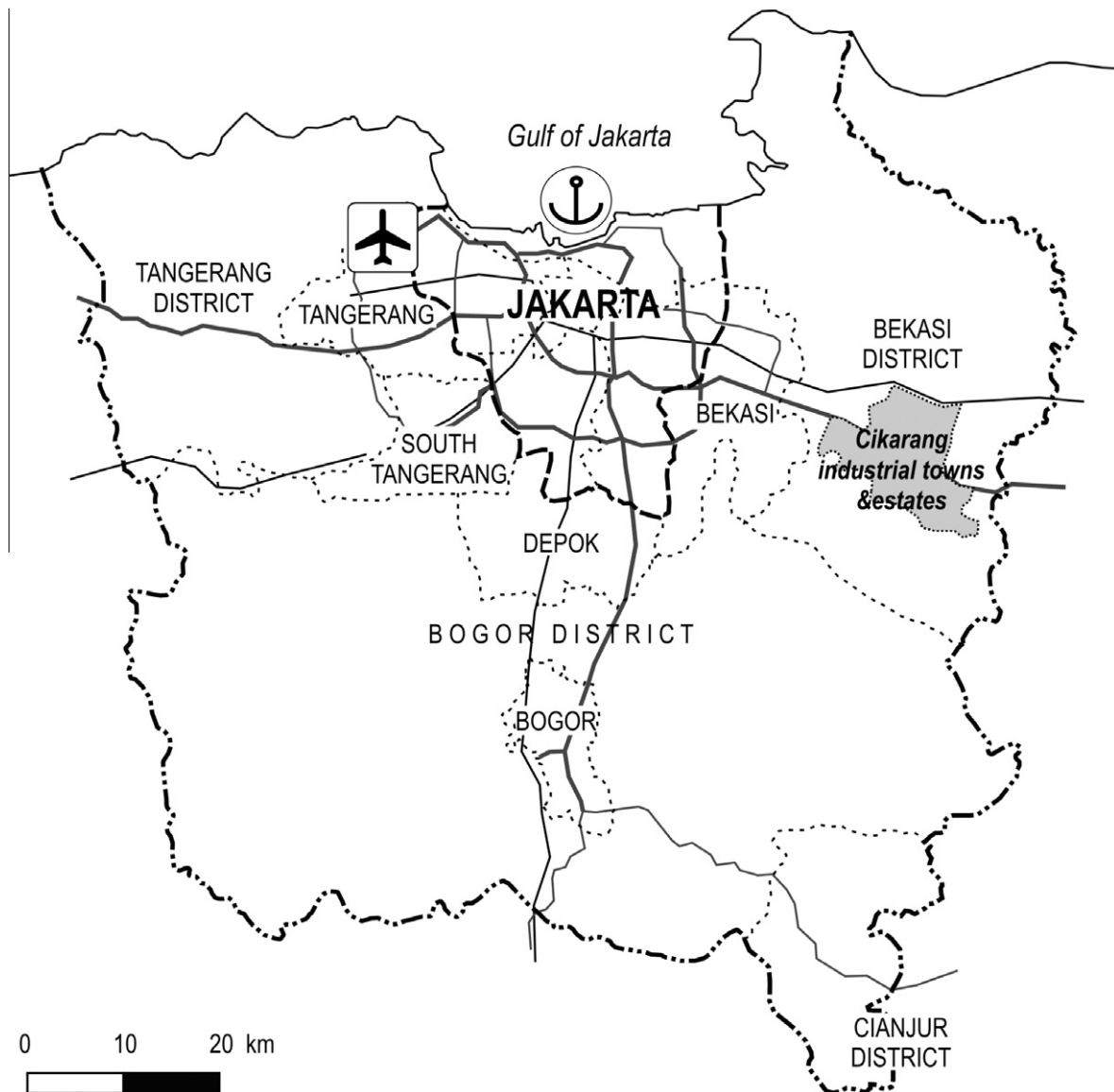


Fig. 1. Map of Jakarta Metropolitan Region (Jabodetabek) showing concentration of industrial estates (including Jababeka) in Cikarang, Bekasi.

outskirts of Jakarta City (Fig. 2). Historical land use maps prepared by Rustiadi (2007) have shown that, up until the beginning of the 1980s, the built-up areas were still concentrated within the established boundaries of Jakarta City. However, since the late 1980s, property development has boomed in the suburbs. During the peak period of 1992–2000, about 90,760 ha of land were converted, typically from agriculture, to built-up areas (Rustiadi, 2007). The 1997 Asia economic crisis abruptly restrained this physical expansion. After the following 3 years of turmoil, the economy gradually recovered and urban expansion continued with the conversion of approximately 22,873.04 ha of land from 2000 to 2005 alone (Rustiadi, 2007).

Suburbanization has in turn affected the economic structure of JMR. Based on the information calculated from the annual data of the Central Bureau of Statistics (Biro Pusat Statistik, 2005) for the years 1985–2005, it can be concluded that tertiary sectors (finance, service and commercial sectors) were still consistently concentrated in Jakarta. On average, the suburbs only contributed 8.7% of the region's GDRP in the finance and service sectors. Meanwhile,

its commercial sectors (including hotels and restaurants) contributed only 22.5% of the region's GDRP in these sectors. Nevertheless, a remarkable shift could be seen in the manufacturing sector. At least in this sector, the suburbs' contribution increased substantially from only 24.6% in 1985–1990 to 59.8% in 2000–2005.

The leading role of the suburbs in the regional economic restructuring is more apparent in the absorption of FDI (foreign direct investment). The annual data provided by the Investment Coordinating Agency (BKPM, 2010) show that, in the last decade, the suburbs have attracted most of the foreign business units in the manufacturing, real estate, and infrastructure sectors (Table 1). From 1998 to 2009, the contribution of the suburbs in these sectors stabilized at 84–87%. In addition, the proportion of FDI in commercial and service sectors in the suburbs was still low but has consistently increased from only 5.4% in 1998–2000 to 9.4% in 2001–2003, 13.0% in 2004–2006, and 14.9% in 2007–2009 (Table 2).

Table 1 also shows that about 50% of the FDI in the secondary sectors in JMR was captured by Kabupaten Bekasi alone. This figure

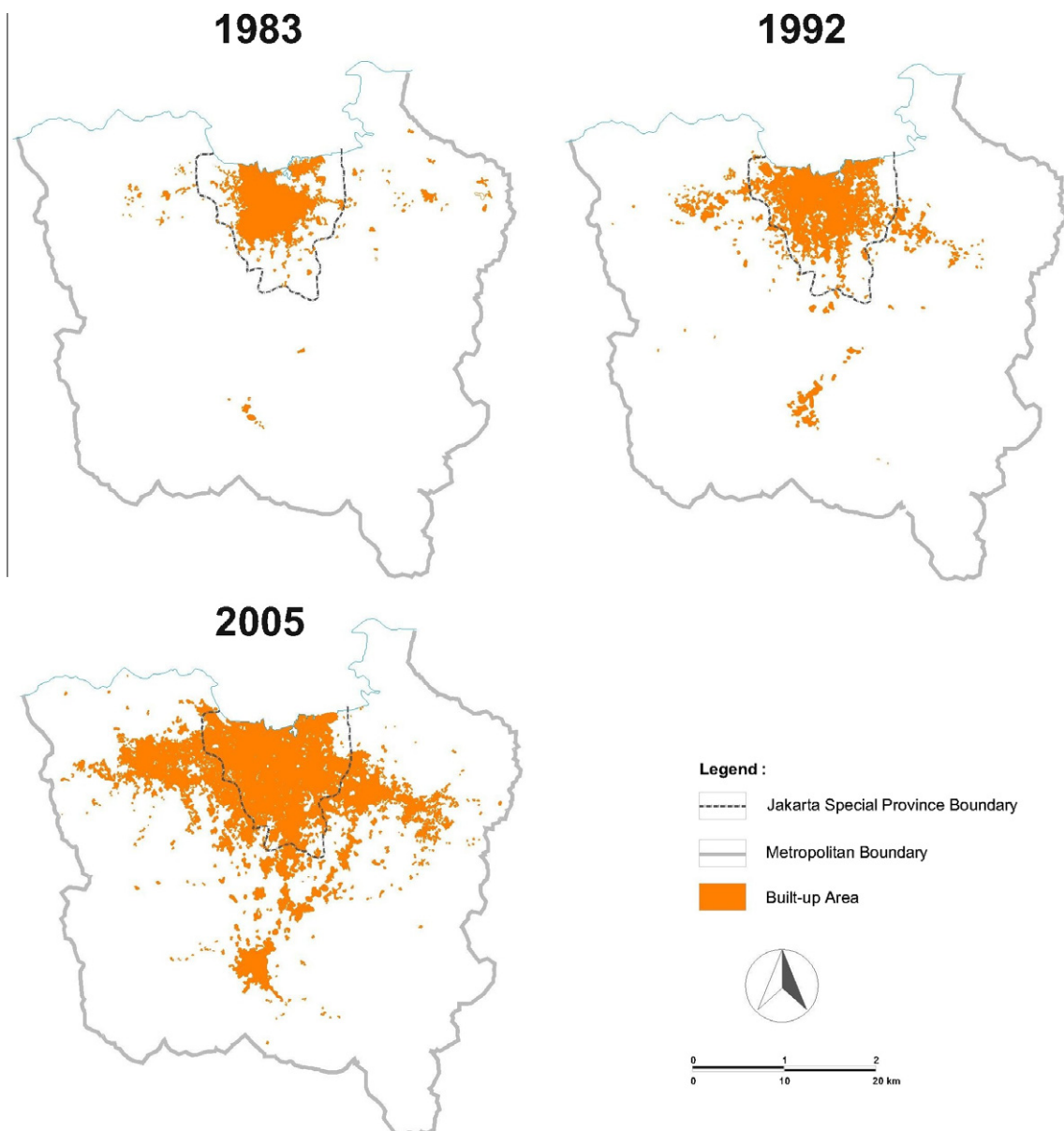


Fig. 2. Expansion of built-up areas in JMR 1983–2005. Source: Analyzed from Rustiadi (2007).

Table 1

FDI in secondary sectors in Jakarta Metropolitan Region. Source: calculated from BKPM (2010).

Region	1998–2000 (%)	2001–2003 (%)	2004–2006 (%)	2007–2009 (%)
Jakarta	15.1	16.2	13.0	12.7
Bodetabek	84.9	83.8	87.0	87.3
Kotas	8.1	7.5	12.3	11.1
Kab. Bogor	16.6	12.7	11.7	10.3
Kab. Bekasi	52.4	51.5	45.0	46.9
Kab. Tangerang	18.1	12.1	18.0	19.0
JMR	100.0	100.0	100.0	100.0

Table 2

FDI in tertiary sectors in Jakarta Metropolitan Region. Source: calculated from BKPM (2010).

Region	1998–2000 (%)	2001–2003 (%)	2004–2006 (%)	2007–2009 (%)
Jakarta	94.6	90.6	87.0	85.1
Bodetabek	5.4	9.4	13.0	14.9
Kotas	1.1	3.2	3.1	3.9
Kab. Bogor	1.1	0.4	1.2	1.4
Kab. Bekasi	2.5	5.0	5.7	7.0
Kab. Tangerang	1.6	0.8	3.1	2.6
JMR	100.0	100.0	100.0	100.0

gradually decreases, but it can be argued that this trend is partly due to the emergence of tertiary sectors in Kabupaten Bekasi. The proportion of Kabupaten Bekasi in receiving FDI in the tertiary sectors has substantially increased from 2.5 in 1998–2000 to 7% in 2007–2009 (Table 2). These figures indicate that Kabupaten Bekasi has stabilized its role as the main industrial center of JMR while taking the lead in the suburban economic restructuring toward service sectors.

Economic and political background and policy contexts

Suburbanization in JMR was triggered by a series of deregulation and debureaucratization measures in the 1980s. Soeharto's market-oriented policy aimed to accelerate economic growth by promoting domestic and foreign private participation in finance and industries. The policy has boosted real estate industry as well as FDI in manufacturing, leading to an uncontrolled growth of large-scale private land development in the suburbs (Firman, 2000). In addition to the influence of global capitalism adopted in Soeharto's market-oriented policy, other institutional factors contributing to JMR's suburbanization include the rise of middle-class society, clientelist governance practices, and the weakened government presence in the suburb (Hudalah et al., 2007). The land subdivision projects mainly took the forms of dormitory and industrial towns located on the outskirts of the city. In the peak period of the 1990s property boom, more than 23 new towns ranging from 500 to 6000 ha were developed around Jakarta alone (Firman, 2004; Winarso & Firman, 2002).

Industrial development played an important role in the suburban transformation of JMR. Progressive government policy with respect to industrial investment actually began after Soeharto took office in the late 1960s. With its market-led development policy, the Soeharto administration emphasized industrialization as a key agenda to boost economic growth. Industrial development was first endorsed in industrial zones (*zona industri*) and later in industrial estates (*kawasan industri*). An industrial zone refers to a concentration of industrial activities that are not necessarily

supported by planned and adequate infrastructure and facilities. Traditionally, local and labor-intensive industries tended to locate in such unplanned areas. Meanwhile, foreign and hi-tech industries preferred to operate in industrial estates.

Kwanda (2000) found that in the early stage of their development in the 1970s, industrial estates were built by the government enterprises as a reaction to increasing environmental impacts and infrastructure inadequacy in industrial zones. However, in response to a growing number of home as well as foreign private investments, through Presidential Decree 53/1989 (Presiden Republik Indonesia, 1989), the government opened the door for private companies to invest and develop industrial estates. After this date, industrial estates grew rapidly. For instance, by 1994, HKI (Himpunan Kawasan Industri Indonesia) or the Indonesia Industrial Estate Association registered 146 industrial estate locations with a total area of 42,019 ha, mostly located on the outskirts of Jakarta – especially in Kabupaten Bekasi – (21,289 ha) and Jakarta (3064 ha) (Kwanda, 2000).

In response to the rapid growth of private industrial estates, Presidential Decree 41 (1996) was issued to set guidelines for industrial estate businesses in Indonesia. The decree defines an industrial estate as “a center for industrial activities with provisions of infrastructure and supporting facilities, which is developed and operated by a licensed industrial estate company”. The industrial estate development was aimed at accelerating industrialization in the regions; facilitating industrial activities; directing industrial location; and strengthening environment-friendly industrialization.

Industrial estate development in Cikarang, Kabupaten Bekasi

Kabupaten Bekasi refers to a suburban region on the eastern outskirts of Jakarta (Fig. 1). Industrial activities, particularly manufacturing, have played a key role in the development of this sub-region. Manufacturing contributes 79.73% of the GDRP of Kabupaten Bekasi (2009). In addition, industrialization was also able to spur the regional economic growth up to 7.42%, far above the national economic growth of 5.6% (Provinsi Jawa Barat, 2009).

To private investors, Kabupaten Bekasi was considered the most appropriate location for private industrial estates (Shahab, 2010). First, it has a direct access to Jakarta–Cikampek Toll Road, the second intercity toll road ever built in Indonesia, which began to operate in 1988. Kabupaten Bekasi is relatively close to the city center of Jakarta (about forty kilometers) and Tanjung Priok International Seaport. Good access to water resources is also important for manufacturing activities, and Kabupaten Bekasi is passed by water directly channeled from Jatiluhur, the biggest reservoir in West Java. Bekasi was one of the national rice cultivation centers. It was considered much cheaper to acquire land from agricultural areas, such as in Bekasi, than to do so from built-up areas in Jakarta. Another important consideration was that the spatial plan of Kabupaten Bekasi was developed to anticipate the development of an industrial sector.

Industrial estates in Kabupaten Bekasi are mostly concentrated in Cikarang, the central part of the sub-region. There are now seven industrial estates companies in Cikarang covering a total area of 14,620 ha (Table 3). Most were established in 1989, when the government started to allow private companies to participate in the development of industrial estates. Four industrial estates, the MM2100, Bekasi Fajar Industrial Estate (BFIE), East Jakarta Industrial Park (EJIP), and Bekasi International Industrial Estate (BIIE), focused their operation on purely industrial development and management. Meanwhile, three other industrial estates (Jababeka, Lippo Cikarang and Pembangunan Deltamas) not only developed industrial areas but also integrated residential and other urban

Table 3

Industrial estates in Cikarang, Bekasi. Source: adjusted from Shahab (2010) and Provinsi Jawa Barat (2010).

Industrial estate/town	Industrial area (ha)	Planning area (ha)	Industrial tenant	Worker
MM2100 and BFIE	1250	2500	440	52,814
EJIP	220	320	87	23,142
BIIE (Hyundai)	140	200	105	27,800
Jababeka	1570	5600	1235	328,510
Lippo (Delta Silicon)	686	3000	378	100,548
Deltamas (Greenland)	^a	3000	43	^a
Total	3866	14,620	2288	532,814

^a No data available.

functions, thus creating “towns” rather than “estates”. In total, 2288 industrial companies are located in the seven industrial estates (and towns), attracting more than a half million workers. Most of the companies are typified by hi-tech industries such as automotive, electronics, chemicals, machine, metal and plastic.

With seven industrial estates alone located in Cikarang, Kabupaten Bekasi has been transformed into the region with the largest industrial estate concentration in Indonesia. West Java Province (2009) calculated that the industrial sector in Kabupaten Bekasi has contributed significantly to national improvements in economic performance. The seven industrial estates (and towns) in Cikarang has a potential export value of up to \$US 15.1–30.56 billion, or about 46% of the national non-oil and gas export of \$US 66.428 billion (2005). This contribution is much larger than that of its runner-up, Batam industrial estate, with \$US 4.6 billion. The government has also successfully extracted taxes totaling 3.4–6 trillion rupiahs from the industrial activities in Cikarang.

In a later development, the expansion of residential and commercial areas was required to support high-tech industrial activities in Cikarang. Both industrial and urban developments have contributed to increase the annual population growth rate of Kabupaten Bekasi by 6–7%. The figure is far above the national rate of 1.25% (Biro Pusat Statistik, 2005). In addition to Indonesian citizens, the data from the Regional Investment Indonesia show that, in 2005, there were also 3004 expatriates living in Kabupaten Bekasi and 6290 others living in Kota Bekasi.

Jababeka City: From industrial park to suburban center

With a total planning area of 5600 ha, Jababeka is the largest manufacturing estate in Cikarang and in all of South East Asia. Jababeka is located 35 km from the Jakarta Central Business District, 55 km from Tanjung Priok Seaport, and 65 km from Soekarno Hatta International Airport (Fig. 1). The industrial estate was developed in 1989 from a vacant and unproductive lot, which was excavated for the production of roof tiles and bricks (Jababeka, 2009). Jababeka currently has more than 1200 tenants ranging from leading multinational to small local companies with more than 300,000 workers. The tenants come from at least 25 countries including the United States, the United Kingdom, France, Germany, the Netherlands, Australia, Japan, China, Taiwan, Singapore, and Malaysia (Provinsi Jawa Barat, 2009).

Jababeka is developed and managed by PT Jababeka tbk, a leading private industrial estate and town developer in Indonesia. The company (Jababeka, 2010) admitted that it has shifted and extended its business areas several times. Triggered by the issuance of presidential decree 53/1989, it began in 1989 as an industrial estate developer. In 1992, the company started to capture the

booming market of property development. The business was even enlarged into township development in 1996. As a further development, in 2007, the company decided to make a radical shift in its core business to go “beyond property” business activities. The later includes several ambitious projects such as the Indonesia Movie-land (36 ha), the Medical City (74 ha), the Cyber City (240 ha), the 130-MW gas-fired power plants and an international dry port (75–150 ha).

To support the industrial operation, Jababeka has developed two clean water treatment plants, two waste water treatment plants, a communication network, and a public transportation network (Jababeka, 2009). Jababeka has also recently developed gas-fired 130-MW power plants. The electricity produced by the power plants is sold not only to tenants in Jababeka's industrial estates but also to other industrial estates in Cikarang-Bekasi and to the National Electricity Corporation (PT PLN). Accommodating new industrial and housing areas, a new toll road interchange is proposed only 3 km away from the existing one.

Jababeka is currently developing an inland port named Cikarang Dry Port. The dry port is built to meet current and future demands, responding to the limited space in the existing main seaport in Jakarta City (Tanjung Priok) and to an increased volume of traffic in the corridor toward the seaport. Built on 75–150 ha of land, the new port is projected to load and unload up to two million TEU (20-foot equivalent units) of containers or two-third of the capacity of Tanjung Priok Seaport. It is expected to be able to facilitate the export–import shipping and logistic activities of the industrial estates and zones in the Jakarta–Bekasi–Cikampek corridor (PT Jababeka Infrastruktur, 2009).

Apart from industrial-oriented infrastructure and facilities, Jababeka is also complete with major urban facilities. These facilities include 30,000 houses, 8 hotels and apartments, 16 universities and institutions of higher education, an international hospital, 24 malls and shopping centers, an international golf course and a botanical garden. With these facilities, Jababeka currently houses more than 958,000 people, including about 2450 expatriates (Provinsi Jawa Barat, 2009).

The Indonesia Movie-land will represent Jababeka's vision of the future of Indonesian TV broadcast and film industry. It will be a fully integrated movie industrial estate using all facilities available in Jababeka industrial city, with 36 ha dedicated for movie industry activities. The movieland is not a theme park, which functions merely to attract tourists. Instead, it is aimed to be a center for creative industrial activities focused on film and television industries. It is envisioned to be “the first one stop film and TV industry center in Indonesia” (Jababeka, 2008). The movieland complex will include TV stations and studio facilities, film production houses, a cultural center (Jababeka Multi-Cultural Center), an ICT training center, accommodations, cafes and entertainment spaces, commercial facilities (including the Hollywood Plaza), residential areas, and integrated internet connections. The ICT training center is expected to regularly train about 1000 workers per year in the fields of ICT network, multimedia and animation. The cultural center is expected to be the nation's first multicultural center. It can seat up to 400 people and feature programs and exhibitions covering a wide range of performances with the goal of providing intellectual and cultural enrichment as well as facilities for the community (Jababeka, 2008).

The Jababeka Medical City is an integrated medical center designed to create a living environment that combines modern urban functions with health and medical care in a countryside setting. The project will include world-class hospital and healthcare facilities, research centers (in cooperation with the Ministry of Research and Technology), laboratories and higher educational institutions (Jababeka, 2009). The whole project site will be linked with a central park, a lake, bicycle and jogging tracks and open green spaces.

The residential complex will include retirement homes and a retirement center.

Socio-economic and spatial implications and governance adaptation

Kabupaten Bekasi is administratively divided into 21 *kecamatan* (subdistricts). Based on their economic structure, the *kecamatan* in Kabupaten Bekasi can be classified into three groups. First, three *kecamatan* in Cikarang are the prime locations of the industrial estates. These central areas are surrounded by 12 suburban *kecamatan*, which serve as the regional industrial hinterland. The last six *kecamatan* tend to be isolated; residents here mostly work in agricultural activities.

The first two groups of *kecamatan* both resemble suburban areas in terms of their population densities (Table 4). Nevertheless, income gaps are obvious between these groups of localities. In 2007, the three industrial *kecamatan* of Cikarang had GDRPs per capita of more than 100 million rupiahs, almost twice as high as that of Jakarta. Meanwhile, the suburban hinterlands and rural *kecamatan* had GDRPs of only a fifth and a tenth of that figure, respectively. The figures imply that the industrial and urban development in Bekasi has grown at the expense of regional cohesion.

This high income inequality has raised security issues among members of the upper- and middle-class society. It has resulted in the congregation of gated communities around the industrial estates, such as in the new towns of Jababeka, Lippo and Deltamas. Some authors have suggested that the socioeconomic segregation in such new towns has reinforced the racial separation of the colonial age (Firman, 2004; Leaf, 1996; Leisch, 2002). In the past, the Dutch colonial government effectively divided the society into three classes based on their ethnic origins: European (the ruler), Indonesian Chinese and *pribumi* (indigenous Indonesian). The first two were very small in proportion, but very powerful in terms of the political realm (European) and economic advancement (Indonesian Chinese). At present, the new towns are places where a considerable number of Indonesian Chinese and expatriates (the modern “ruler”) live voluntarily. Every new town mostly consists of clusters of residential neighborhoods surrounded by security guards and/or concrete walls.

At the local level, the societal divide is reflected in the physical fragmentation between the industrial estates. There has been a lack of communication between the industrial estate companies. In fact, several industrial estates have been specifically developed to accommodate groups of investors from particular countries of origin, such as Japanese (MM2100, BFIE, and EJIP) and Korean (BIIE).

From a neoclassical theoretic viewpoint, a large concentration of industries in one location may create economics of scale, thus increasing economic efficiency through, for example, sharing responsibilities of infrastructure provision. Interestingly, in Cikarang, this sharing has not been the case. Each industrial estate built its own urban infrastructures and facilities, such as a road network,

telecommunication network, wastewater treatment plant, and clean water treatment plant, without clear coordination. As a result, the infrastructures built by different estates tend to be disconnected from each other. Such fragmented infrastructure networks are an unavoidable outcome of the trend toward privatization and liberalization of urban development (Graham & Marvin, 2001).

Rethinking the governance structure

Apart from the socioeconomic and spatial implications at the local and regional scales, since the 2000s, the industrial development in Indonesia, including in Cikarang, faces a new challenge. There are now more emerging East Asian countries that have developed industrial estates to attract FDI. In this context of tighter global competition, the industrial estates are forced to improve their performance. Therefore in 2006, the seven industrial estates in Bekasi signed a Memorandum of Understanding (MOU) witnessed by the Department of Public Works, the Government of the West Java Province, the Government of Kabupaten Bekasi, PT Jasa Marga (the Indonesia Highway Corporation) and other developers in the Cikarang. The purpose of the MOU was to improve services within the industrial community in Cikarang-Bekasi, thus allowing investors to do business with them more conveniently (ZONI, 2007). The seven industrial estates agreed to cooperate and form a special economic zone named ZONI (*Zona Internasional* or International Zone). Through this unification, the industrial estates in Cikarang can better coordinate with the government to create an integrated plan, and thus infrastructure development can be done more efficiently and at a lower cost. According to the MOU, the industrial estates were expected to carry out major infrastructure programs including among others the construction of bridges and flyovers, the construction of additional lanes in the Jakarta-Cikampek toll road, and the expansion of the roads in the vicinity of Cikarang region (ZONI, 2007).

Learning from China, Vietnam, Thailand and India, where the economy is highly influenced by the development of special economic zones, it was argued that the government bureaucracy needed to focus on investment facilitation and the improvement of the business climate (Provinsi Jawa Barat, 2009). Therefore, as a response to the MOU, the government of West Java Province plans to institutionalize the special economic zone proposed by the private sector into Presidential Instruction 3/2006 on fostering investment and economic growth in West Java. The governor formed a coordinating team for infrastructure development in the industrial estates of Cikarang (2007) and prepared a detailed plan for the special economic zone in Cikarang Bekasi (2009). The provincial government (Provinsi Jawa Barat, 2009) defined the Special Economic Zone or *Kawasan Ekonomi Khusus* (KEK) as a designated area in which special regulations are applied to custom duties, taxes, licenses, immigration and labor. The government emphasized that the operation of KEK must be supported with infrastructure and professional management institutions meeting international standards. It was expected that the development of

Table 4
Physical and socio-economic figures for Kabupaten Bekasi and Jakarta. Source: calculated from Kabupaten Bekasi (2009).

Sub-region (subdistrict)	Area size (km ²)	Population	Density (/km ²)	GDRP (million Rp)	PCI (Rp)
Kab. Bekasi	1264	2,125,960	1682	73,939,991	34,779,578
Industrial estates (3)	139	416,520	3002	42,886,825	102,964,624
South Cikarang	52	85,260	1648	13,400,818	157,175,905
North Cikarang	43	168,181	3884	14,575,850	86,667,637
West Cikarang	44	163,079	3733	14,910,158	91,429,048
Suburbs (12)	537	1,196,322	2229	26,389,931	22,059,221
Rural areas (6)	588	513,118	872	4,663,235	9,088,036
Jakarta	662	9,588,198	14,494	567,706,984	59,208,934

a special economic zone in Cikarang would be able to retain the existing investors in manufacturing as a footloose industry and to invite new foreign investments, expand access to global market, stimulate local industries, and create more job opportunities (Provinsi Jawa Barat, 2009).

The end of desakota regions?

In the late 20th century, East Asia's regional-scale urbanization was characterized by the emergence of desakotas, referring to distinct zones between cities and their rural hinterlands. Desakota implied a gradual transformation of previously fertile agricultural areas into built-up areas due to the pressure of urban expansion from the metropolitan core. Desakotas represented the most dynamic part of a metropolitan region that developed incidentally along the main corridors between cities (McGee, Ginsburg, Koppel, & McGee, 1991). Unplanned industrial development in the rice fields was one of common elements of desakota regions (Leaf, 1996).

Moving away from this traditional concept, which mainly views urban transformation on the outskirts of East Asian large-cities as gradual, incidental and unplanned, this article shows that the spatial pattern of JMR might signify the early stages of what in Western terms is called post-suburbia. In the context of JMR, post-suburbia involves a deconcentration of hi-tech industries and multinational companies, converting formerly neglected rural hinterland into planned suburban industrial estates. Furthermore, several suburban industrial estates, such as those found in Cikarang, Bekasi, are now being transformed into new urban centers completed with major commercial, leisure and cultural facilities with a strong Western flavor.

In the beginning, the post-suburban transformation in JMR could be seen as the spatial implication of the regional economic restructuring from agricultural to manufacturing sectors. In the future, it seems that the domination of the manufacturing sectors will be gradually followed by service sectors. Ambitious projects beyond land development and property business such as Movie-land (considered to be the Hollywood of Indonesia), the Medical City, and the Cyber City in Cikarang may further transform the suburb from merely industrial estates toward the region's global cultural center.

The continuity of growth and integration with global capitalism, coupled with decentralizing governance attributes, have made post-suburban development an inevitable phenomenon in JMR. However, given the specific historical context, post-suburbia carries its own meaning in the JMR, which are represented in the unique features of suburban transformation in this particular region.

First, the carrying capacity of the metropolitan core of Jakarta has reached its saturation point, and thus Jakarta alone can no longer cope with tighter global competition in terms of attracting FDI. Land scarcity, infrastructure deficiency, and other factors contributing to the higher production costs in the city have forced new FDIs, especially in manufacturing, to locate in the suburbs. It can be concluded that the spatial transformation in the suburb is an inevitable consequence of the inability of the global city to accommodate further growth. In other words, post-suburban transformation in JMR was created by opportunity rather than by creativity.

Furthermore, post-suburbia in JMR is not directly designed by the government. Yet, it is also not totally improvised. It can be argued that the government still played an indirect role by enacting principal regulations, which have proven to boost property and industrial development. Nevertheless, it was the private companies' own decisions to carry out the development and to locate on the edge of metropolis. A more direct role of the government might be apparent in the future, as recently shown by the provin-

cial government's plan to promote the industrial estates in Cikarang as a special economic zone (Provinsi Jawa Barat, 2009, 2010). However, the plan was not built on clear entrepreneurial ambition and long-term vision, but rather was a reaction to the threat of declining global competitiveness of the suburban industrial estates.

Furthermore, in the United States, post-suburbanites tend to be seen as opportunists seeking maximum individual freedom and limited planning rules (Garreau, 1991; Kling et al., 1991b). Such a libertarian viewpoint might not be entirely relevant in the case of JMR. The middle- and upper-class society and the hi-tech industrial economy in JMR need consistency in planning rules, at least for their own security and quality of life. The need for consistent planning rules was unable to be met in the city because the city government in Indonesia was not equipped with the required institutional capacity to realize agreed planning rules (Hudalah et al., 2007). As a result, the middle class population moved to suburbs where private developers are capable of designing as well as implementing planning rules, thus creating a more secure and livable urban environment.

Post-suburban transformation has largely contributed to the improvement of the economic performance of JMR. However, major sustainability issues have also arisen alongside this transformation. The issues include, among others, regional imbalance, infrastructure mismatch, physical fragmentation, social segregation, and environmental degradation. These issues imply the necessity for more innovative policy instruments, better planning coordination at the regional level, and an increased capacity for local government.

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